Chest X-ray Diagnosis with Joint Report Generation and Lesion Localization Using Multimodal Vision–Language Models

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2025 IEEE 8th International Conference on Pattern Recognition and Artificial Intelligence (PRAI) | 979-8-3315-7405-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/PRAI67447.2025.11412806

***Abstract*—Lung diseases remain a major global health concern, and chest X-ray imaging is a critical tool for early diagnosis and treatment. To support radiologists, automatic computer-aided diagnosis (CAD) systems must detect abnormalities and accurately localize them within the image. However, developing such systems faces key challenges: annotated bounding box labels are scarce, and many models lack clear interpretability and visual grounding to support clinical trust. To address these gaps, we propose a unified multimodal diagnostic framework that combines a DenseNet-121 classifier with Grad-CAM-based pseudo-labeling and a powerful vision-language model (Qwen2.5-VL-7B-Instruct). The DenseNet classifier is trained on pneumonia and pneumothorax cases from the MIMIC-CXR-JPG dataset, with Grad-CAM used to extract pseudo bounding boxes by thresholding activation maps. These pseudo-labels, combined with limited expert annotations, supervise the fine-tuning of Qwen2.5-VL, enabling it to generate structured radiology reports alongside spatial coordinates for pathological findings. Experimental results demonstrate that incorporating pseudo-labeled data significantly boosts localization performance, achieving 26.09% recall at IoU $\geq$ 0.25 compared to 4.35% using real labels alone. Meanwhile, report generation quality remains strong, with BLEU-4 and ROUGE-L scores comparable to state-of-the-art systems and with faster inference than models like LLaVA. This work illustrates how Grad-CAM-based pseudo-labeling and multimodal vision-language models can enhance interpretability and efficiency, providing a practical pathway toward integrated, explainable AI for chest X-ray diagnosis.**

***Index Terms*—chest X-ray, vision–language models, radiology report generation, lesion localization, weak supervision**

I. Introduction

Chest radiography is one of the most common and essential imaging examinations in medicine, and it is used to diagnose various cardiopulmonary conditions. However, interpreting chest X-rays is time-consuming and requires significant expertise. The growing volume of imaging studies can overwhelm radiologists, leading to delays and the risk of missed findings [1]. This has motivated extensive research into automated chest X-ray analysis, including algorithms for detecting pathologies and even generating complete radiology reports. Early work in this domain, such as CheXNet [2], focused on image classification or object detection for specific abnormalities (e.g., detecting pneumonia) using convolutional neural networks. These methods demonstrated the utility of MIMIC-CXR-JPG, a large public dataset derived from the broader MIMIC-CXR database, for training models to identify common findings or predict radiological labels directly from medical images. MIMIC-CXR is a vast, de-identified public resource comprising chest radiographs and their corresponding free-text reports [3].

Moving beyond classification, researchers have explored chest X-ray report generation by framing it as an image-to-text translation task. Traditional image captioning approaches have proven inadequate, since radiology reports are typically long, detailed, and filled with specialized medical terminology [4]. More advanced methods, such as encoder–decoder architectures with attention and memory modules, were among the first to show promising results for automatic report generation on large-scale datasets like MIMIC-CXR [5]. Despite these advances, a key limitation of most earlier systems is the lack of visual grounding: they generate free-text descriptions of findings without indicating where each finding is located within the image. Moreover, traditional deep learning methods for medical report generation often struggle to maintain consistency and accuracy, especially when dealing with subtle or rare findings. This underscores the need for approaches that more effectively align textual descriptions with visual evidence while making efficient use of limited annotated data.

Multimodal vision–language models (VLMs) are creating new opportunities for generating higher-quality medical reports. Recent progress in this field has shown that models such as PaLM-E [6] and Flamingo [7] can jointly process visual and textual information, delivering flexible outputs beyond simple image captioning, including medical report generation and image understanding tasks. In particular, instruction-tuned VLMs—which accept images and follow natural-language prompts—are especially well-suited for complex medical applications requiring integrated visual and linguistic reasoning. A notable example is Med-PaLMM [8], a generalist biomedical VLM created by fine-tuning PaLM-E on the MultiMed-

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Bench benchmark. It achieves good performance on chest X-ray report generation and other clinical vision–language tasks. However, even these advanced VLMs typically produce good free-text outputs. The challenge of generating structured outputs, such as bounding-box annotations or other visual markers that localize findings within images, remains largely unexplored in the medical setting.

Recent work has shown that bounding-box detection in chest X-rays is feasible when fully supervised annotations are available. For example, Khalili et al. demonstrated effective localization and classification of chest X-ray abnormalities using a fully supervised Mamba–YOLOvX detection framework [9]. However, the major limitation remains: most bounding box–annotated chest X-ray datasets contain far fewer than the total number of available images—e.g., MIMIC-CXR includes over 300,000 images, yet only about 400 have expert-drawn bounding boxes [10]. This label scarcity restricts large-scale fully supervised training and motivates using pseudo-labeling or weakly supervised methods to scale lesion localization more efficiently.

In this work, we propose a unified system for chest X-ray diagnosis that generates a complete radiology report and simultaneously provides the location of key abnormalities. Our system is built on a state-of-the-art multimodal VLM (Qwen2.5-VL-7B [11]), which we fine-tune to produce radiology reports with integrated visual localization. To overcome the scarcity of localization annotations for training, we employ a weakly supervised bounding box generation strategy: we train a CNN classifier for certain pathologies and use its Grad-CAM [12] heatmaps to create pseudo bounding box labels for those findings. These pseudo-labeled examples are then used to augment the small set of human-annotated examples during fine-tuning. By combining text and image supervision in this way, the model learns to associate the report's content with specific regions in the image. In summary, the main contributions of this work are as follows:

- We develop a unified system for chest X-ray diagnosis that jointly generates radiology reports and localizes lesions, improving interpretability and clinical relevance.
- We address the lack of bounding-box annotations by combining expert-drawn labels with pseudo bounding boxes derived from Grad-CAM, enabling effective weakly supervised learning.
- We validate our system on the large-scale MIMIC-CXR-JPG dataset, showing that combining pseudo-labeled and real annotations substantially boosts localization recall (from $\sim$4.35% to $\sim$26.09% at IoU $\geq$ 0.25) while maintaining strong report generation quality.

II. Related Work

A. Automatic Radiology Report Generation

Traditional methods for automatic radiology report generation have largely adapted image captioning or encoder–decoder frameworks developed for general vision–language tasks. While these approaches laid the groundwork, they often produce generic, repetitive, or templated text that fails to reflect the complexity and specificity needed in real clinical reports. For example, Jing et al. [13] combined a co-attention mechanism with hierarchical LSTMs to better capture sentence-level dependencies in chest X-ray reports. Similarly, Chen et al. [14] introduced a Transformer architecture augmented with memory modules to enhance long-range coherence and medical terminology usage. However, these conventional pipelines still struggle to detect subtle or rare findings and often lack the precision required for accurate clinical decision-making.

B. Vision-Language Models in Medical Imaging

Recent advances in vision-language models (VLMs) have enabled more flexible and context-aware medical image interpretation by combining visual understanding and language generation in a unified framework. For example, Med-PaLMM [8] extends the generalist PaLM-E model through domain-specific fine-tuning on the MultiMedBench benchmark, achieving strong results on chest X-ray report generation and other clinical image-text tasks. Similarly, LiteGPT [15] demonstrates that a lightweight VLM architecture with explicit bounding-box supervision can jointly handle classification and localization for chest X-rays, paving the way for more structured and interpretable outputs. However, while some large VLMs can perform localization, many still focus mainly on generating free-text reports without providing precise spatial references for abnormalities, which limits their usefulness for downstream tasks like lesion examination or treatment planning. This motivates the need to integrate robust localization capabilities into VLMs for medical imaging.

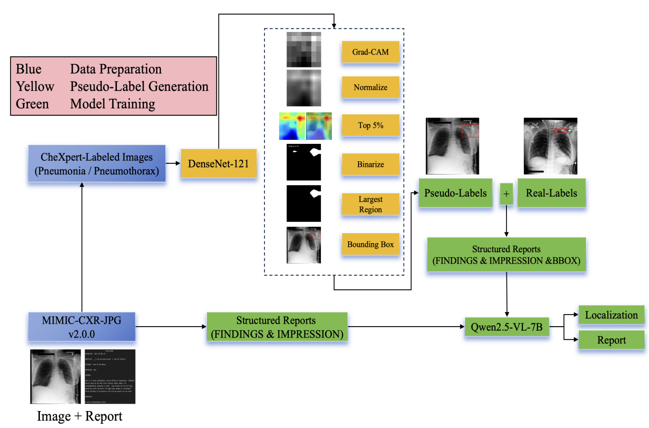

Fig. 1: Overview of our proposed approach.

C. Lesion Localization and Weak Supervision

Lesion localization is critical in medical imaging, as it allows clinicians to pinpoint abnormal regions for further examination, diagnosis, or intervention [16]–[18]. Accurate localization typically requires detailed pixel-level or bounding-box annotations, which are expensive and time-consuming to obtain at scale. To address this, many studies have explored weakly supervised approaches that use only image-level labels or partial annotations to localize abnormalities. The most widely used approach for localization without bounding box annotations is the use of class activation maps (CAMs) [12], [19], [20]. Wang et al. [21] introduced ChestX-ray8, a large-scale dataset with image-level labels for common thoracic diseases and demonstrated how weak supervision can enable both classification and localization. More recent work, such as Wang et al. [22], improves weakly supervised localization by applying non-linear modulation and foreground control to distinguish relevant regions better. Anatomy-guided strategies have also been proposed; Yu et al. [23] showed that leveraging anatomical structures can further boost weakly supervised abnormality localization in chest X-rays. These approaches highlight that effective localization remains feasible even when detailed annotations are limited, supporting the development of trustworthy and scalable clinical AI systems.

III. Methodology

As shown in Figure 1, our method fine-tunes the Qwen-2.5 VL 7B vision–language model to simultaneously generate text reports and detect abnormalities. First, it leverages the strong pretrained knowledge of a large VLM (Qwen-2.5 VL 7B), ensuring fluent report generation and robust visual understanding even with limited fine-tuning data. Second, to address the challenge of having only about 400 labeled samples, we train a DenseNet for image-level classification (e.g., "pneumonia present") and apply Grad-CAM to generate foreground heatmaps, which we then use to create pseudo bounding-box labels. The detection component is trained using both real and pseudo labels. Third, our model produces structured outputs, including a standard radiology report with `Findings` and `Impression` sections and machine-readable coordinates for any localized lesions. This structured output can facilitate integration into clinical workflows, for example, by highlighting detected findings within a PACS (Picture Archiving and Communication System) or providing more transparent AI decision support.

A. Dataset Overview

We conduct our experiments using two main subsets derived from the MIMIC-CXR-JPG corpus. For report generation, we select a subset that pairs chest X-ray images with their corresponding radiology reports. For localization, we use another subset along with the expert-annotated bounding boxes provided by Tam et al. [10], which include 458 radiologist-drawn annotations for pneumonia and pneumothorax cases across approximately 400 images. Details of each subset are provided in Section III-B.

Report Generation Data. For the multimodal report generation task, we structure the training samples as multi-turn conversations. Each example includes an image and a prompt instructing the model to generate a report with `FINDINGS` and `IMPRESSION` sections. The ground truth response is the corresponding radiologist-written report. These samples follow the format:

```
{
  "conversations": [
    {"from": "system", "value": "You are a
    professional radiologist..."},
    {"from": "human", "value": "<image>Give a
    short report..."},
    {"from": "gpt", "value": "FINDINGS:...\
    nIMPRESSION:..."}
  ],
  "images": ["/path/to/image.jpg"]
}
```

We extract the `FINDINGS` and `IMPRESSION` sections from these reports for training the multimodal models.

Real Label for Localization. We selected 300 images for training and 54 for evaluation from the annotated subset. Each sample is paired with a radiology report and ends with a structured bounding box entry, formatted as:

```
{
  "conversations": [
    {"from": "system", "value": "You are a
    professional radiologist..."},
    {"from": "human", "value": "<image>Give a
    short report..."},
    {"from": "gpt", "value": "FINDINGS:...\
    nIMPRESSION:...\n[BBOX] Right lower lobe
    pneumonia 29.29 115.55 95.27 190.46"}
  ],
  "images": ["/path/to/image.jpg"]
}
```

This label format provides both the textual finding and its spatial location, which serve as strong supervision for training the model to generate both narrative and bounding box outputs.

B. Model Training

We design two separate training pipelines for distinct objectives: (1) pure radiology report generation, and (2) joint report generation with lesion localization. The first pipeline is designed to compare medical report generation across different vision–language models.

1) Report-Only Generation: To compare baseline performance in medical report generation, we fine-tune two large vision–language models: **LLaVA-1.5-7B** and **Qwen2.5-VL-7B-Instruct**. Both models are trained using identical configurations to ensure fair comparison.

Training Configuration:

- Training method: Supervised fine-tuning (SFT) with LoRA (rank = 8)
- Image resolution: 256×256
- Max text length: 2048 tokens
- Learning rate: 1e-4
- Epochs: 15
- Batch size: 4 (2 per device with gradient accumulation)
- Precision: bf16 mixed precision
- Hardware: $4 \times$ NVIDIA RTX 3090 GPUs

Training Data: Based on our experimental setup, we select a total of 21,907 radiograph–report pairs from the MIMIC-CXR dataset. Each sample consists of a frontal-view chest X-ray and a corresponding structured report containing `FINDINGS` and `IMPRESSION` sections. No bounding boxes are included in this configuration.

2) Report + Lesion Localization: For the joint task of report generation and abnormality localization, we fine-tune **Qwen2.5-VL-7B-Instruct**, which natively supports both text generation and bounding-box coordinate prediction. To improve performance given the limited number of bounding-box annotations, we generate pseudo labels and train the model using a combination of real and pseudo-labeled data.

Pseudo-Label Generation. To scale localization training, we generate pseudo bounding box annotations using Grad-CAM. We first train a DenseNet-121 [24] classifier using 8,161 chest X-ray images labeled with either pneumonia or pneumothorax, selected from the MIMIC-CXR-JPG dataset. Since our 458 expert-annotated bounding boxes are predominantly for pneumonia and pneumothorax, we restrict the classifier's training set to these two conditions to maintain task consistency. The class-wise distribution is as follows:

- Training: 4191 pneumonia, 2326 pneumothorax (total 6517).
- Validation: 519 pneumonia, 308 pneumothorax (total 827).
- Test: 503 pneumonia, 314 pneumothorax (total 817).

Once the classifier is trained, we apply it to a separate, non-overlapping set of 1,680 high-confidence pneumonia and pneumothorax images—also sampled from MIMIC-CXR-JPG but excluded from classifier training and validation. This subset is independently used for pseudo-label generation. For each image, we compute Grad-CAM heatmaps corresponding to the predicted class.

The pseudo-labeling process involves running a DenseNet-121 classifier on each image to obtain class predictions, followed by Grad-CAM to extract attention maps. These maps are resized to 256×256, normalized, and thresholded to retain the top 5% activation values. A binary mask is generated, from which connected components are identified. The largest region is selected, and its bounding box is extracted and used as a pseudo-label. Each resulting training sample is constructed by appending the Grad-CAM-generated bounding box to the end of the corresponding real radiology report, which contains `FINDINGS` and `IMPRESSION` sections. Since the pseudo-labeling process only produces spatial coordinates without textual disease names, the final format includes a bounding box but no explicit lesion label:

```
[BBOX] x_min y_min x_max y_max
```

These are treated as pseudo labels and used to augment the training of our multimodal model.

When training the joint task of report generation and abnormality localization, we also compare three different settings: using only real labels, using a combination of real and pseudo labels, and using only pseudo labels.

Training Variants:

- **Real-Only:** 300 samples with human-annotated bounding boxes.
- **Fake-Only:** 1680 pseudo-labeled samples generated via Grad-CAM.
- **Real+Fake:** 300 real samples combined with 300, 600, 900, 1200, or 1500 pseudo-labeled samples.

These configurations are designed to evaluate how increasing pseudo-label quantity influences localization performance while preserving report quality.

Training Configuration:

- Training method: Supervised fine-tuning (SFT) with LoRA (rank = 8)
- Image resolution: 256×256
- Max text length: 2048 tokens
- Learning rate: 5e-5
- Epochs: 30
- Batch size: 2 (1 per device with gradient accumulation)
- Precision: bf16 mixed precision
- Hardware: 4 $\times$ NVIDIA RTX 3090 GPUs

Data Format: Each training sample consists of a radiology report and a final line indicating the lesion location in normalized coordinates:

```
FINDINGS: ...
IMPRESSION: ...
[BBOX] x_min y_min x_max y_max
```

No explicit class label is attached to the bounding box, and both real and fake labels follow this structure.

IV. Results and Evaluation

We evaluate our system from two perspectives: (1) the quality of the generated radiology reports using a pure text generation task, and (2) the accuracy of lesion localization using bounding boxes.

A. Report Generation Quality (Text-Only Task)

We evaluate the performance of Qwen2.5-VL-7B-Instruct and LLaVA-1.5-7B on the task of radiology report generation using both qualitative and quantitative comparisons.

Table I presents automated language metrics on 3,000 radiograph-report pairs sampled from the MIMIC-CXR dataset, using the same experimental settings. Both models achieve comparable scores across BLEU-4, ROUGE-1, ROUGE-2, and ROUGE-L, with Qwen showing a slight edge in BLEU-4 and ROUGE-2.

In addition to these metrics, Table II compares inference efficiency. Qwen2.5-VL-7B achieves significantly higher throughput and faster runtime than LLaVA-1.5-7B, demonstrating its advantage for large-scale deployment.

Finally, Table III provides a qualitative comparison between ground-truth reports and model-generated outputs. Qwen2.5-VL and LLaVA-1.5 both produce coherent, clinically relevant reports, but differences in findings and impressions highlight the importance of high-fidelity text generation for accurate clinical decision-making.

Overall, while the two models deliver similar text quality, Qwen2.5-VL-7B offers a clear benefit in inference speed and efficiency, making it a strong choice for real-world medical reporting pipelines.

TABLE I: Automated language evaluation metrics on the test set reports. Qwen2.5-VL-Instruct and LLaVA-1.5-7B demonstrate comparable performance.

Model	BLEU-4	ROUGE-1	ROUGE-2	ROUGE-L
Qwen2.5-VL	38.65	43.42	24.99	36.24
LLaVA-1.5	38.41	43.44	24.98	36.31

TABLE II: Inference performance comparison between Qwen and LLaVA models. Qwen achieves significantly higher throughput and faster runtime.

Metric	LLaVA	Qwen	Better
Runtime	9m49s	5m08s	Qwen
Samples/sec	5.086	9.723	Qwen
Steps/sec	0.080	0.152	Qwen

B. *Localization Accuracy (Text + BBox Task)*

We evaluate lesion localization performance on 54 test chest X-rays with expert-annotated ground-truth bounding boxes. To quantify localization quality, we report recall at three Intersection-over-Union (IoU) thresholds: 0.1 (lenient), 0.25 (moderate), and 0.5 (strict).

Table IV summarizes recall results across different training configurations. Models trained with both real labels and Grad-CAM-generated pseudo-labels consistently outperform those trained with real labels alone or only pseudo-labels. For example, using 900 pseudo-labeled samples combined with 300 real labels achieves a recall of 26.09% at IoU>0.25, a substantial improvement over the 4.35% recall with real labels alone. We also observe that adding more pseudo-labeled samples generally improves performance up to 1500 samples, with diminishing returns beyond that point.

Table V and Figure 2 provide a qualitative perspective on this trend. Table V shows IoU scores for three representative chest X-ray images under three training scenarios: Real Only, Fake Only, and Real + Fake. The Real + Fake model achieves the highest score in Table IV, specifically the configuration using 1500 pseudo-labeled samples. The combined Real + Fake model achieves significantly higher IoUs, especially for images where real supervision alone or pseudo-labels alone fail to localize abnormalities. Figure 2 visually illustrates these differences, comparing predicted bounding boxes from each model against the ground truth. The rightmost column shows how combining real and pseudo labels results in more accurate and complete localization.

Overall, these results demonstrate that integrating pseudo-labels with a small number of real annotations effectively improves lesion localization accuracy. This finding underscores the benefit of weakly supervised approaches for visual grounding, especially when high-quality annotations are scarce.

V. Discussion

Our findings demonstrate the feasibility of a unified vision–language system that can simultaneously generate comprehensive radiology reports and localize lesions on chest X-rays. This integrated approach marks an essential step toward more interpretable and clinically trustworthy AI tools in medical imaging. By combining textual and spatial outputs, our system allows radiologists to read what the AI reports and verify where the abnormality is detected in the image, fostering greater transparency and user confidence.

A key takeaway is the practical benefit of weakly supervised localization through Grad-CAM-generated pseudo-labels. While these pseudo bounding boxes are inherently coarse, they significantly enhance the model's ability to align

TABLE III: Qualitative comparison between ground-truth and model-generated reports.

Ground-Truth Report
FINDINGS: Emphysema is severe. Calcific left fibrothorax restricts the volume of the left lung and obscures its structure. There may be a nodule in the right lung at the level of the anterior right fifth rib. A small pneumonia would likely be missed, detectable only by CT scanning. Heart size is normal and there is no pulmonary edema. **IMPRESSION:** Severe emphysema. Restrictive calcific left fibrothorax. Possible left lung nodule. Any small pneumonia unrecognizable. Chest CT suggested.
Qwen2.5-VL Output
FINDINGS: There has been interval removal of the enteric tube and left IJ central line. There is a small left apical pneumothorax. There is a small left pleural effusion. Left lower lobe atelectasis is unchanged. There is no focal consolidation concerning for pneumonia. The cardiomediastinal and hilar contours are stable demonstrating mild cardiomegaly. There is no pleural effusion or pneumothorax on the right. No acute osseous abnormalities identified. **IMPRESSION:** Small left apical pneumothorax. Small left pleural effusion. Left lower lobe atelectasis.
LLaVA-1.5 Output
FINDINGS: Frontal and lateral views of the chest demonstrate increased opacification of the left costophrenic angle and left lung base, consistent with a layering pleural effusion. The right lung is clear. There is no pneumothorax. Mild cardiomegaly is stable. The mediastinal and hilar contours are normal. **IMPRESSION:** Left pleural effusion. Stable mild cardiomegaly.

TABLE IV: Lesion localization recall at different IoU thresholds for models trained with 300 real labels and varying numbers of pseudo-labeled samples. Evaluation is conducted on 54 test images with ground-truth annotations.

Training Data	IoU>0.1	IoU>0.25	IoU>0.5
Real Only	14.49%	4.35%	0.00%
Real + Fake 300	23.19%	14.49%	2.90%
Real + Fake 600	18.84%	11.59%	1.45%
Real + Fake 900	30.43%	26.09%	5.80%
Real + Fake 1200	27.54%	18.84%	8.70%
Real + Fake 1500	34.78%	24.64%	7.25%
Fake Only (1680)	4.35%	4.35%	0.00%

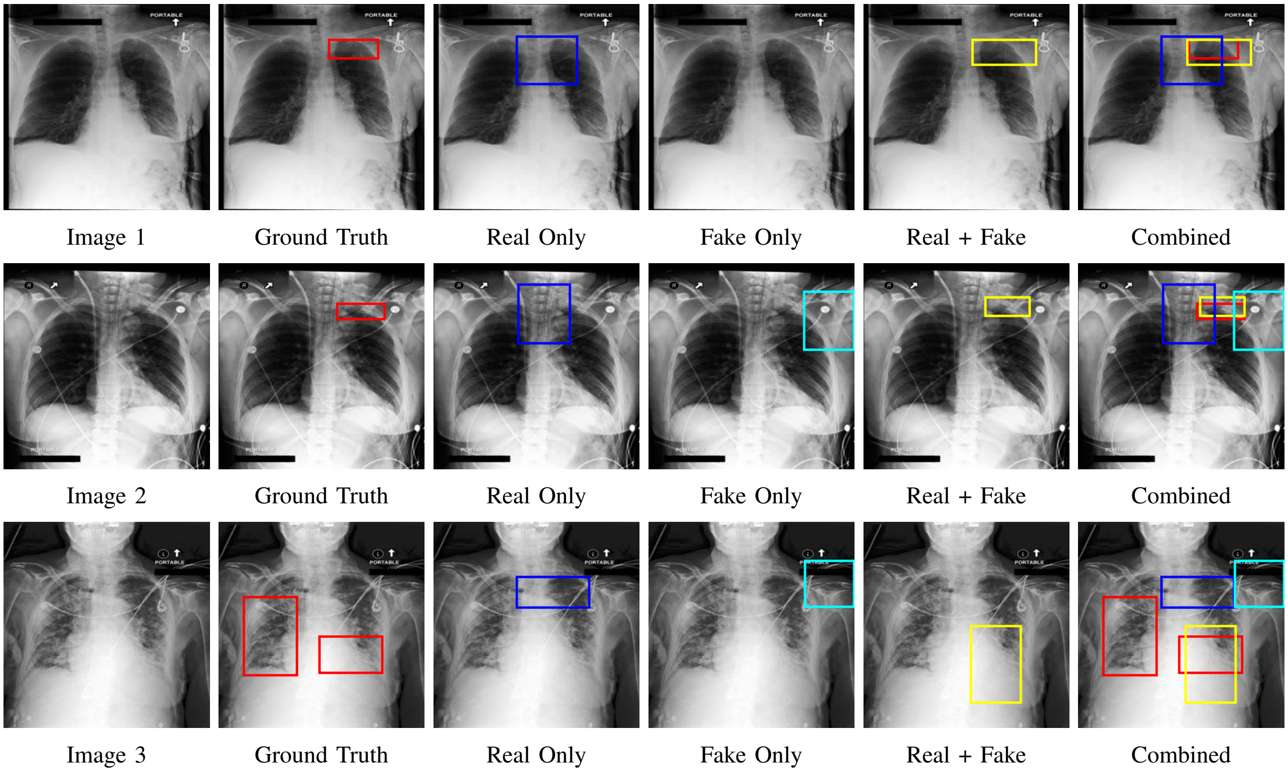

Fig. 2: Qualitative comparison of bounding box predictions across models on three sample chest X-rays. **Red** = Ground Truth; **Yellow** = Real + Fake prediction; **Blue** = Real Only prediction; **Cyan** = Fake Only prediction. Rightmost column shows all bounding boxes combined.

TABLE V: IoU comparison for three sample chest X-ray images across different models.

Image	Real Only IoU	Fake Only IoU	Real + Fake IoU
Image 1	0.213	0.000	0.599
Image 2	0.088	0.056	0.536
Image 3	0.000	0.000	0.434

descriptive text with the relevant image regions. For example, when the model outputs “right lower lobe pneumonia,” pseudo-label training also encourages it to generate a bounding box over the right lower lung zone, creating explicit visual grounding. This connection between language and spatial evidence is vital for real-world acceptance. Without it, a report could be technically correct but clinically unverifiable.

Our design demonstrates an effective knowledge transfer: we leverage a simple DenseNet classifier and its Grad-CAM heatmaps to generate pseudo-labels, guiding fine-tuning a large multimodal VLM (Qwen2.5-VL). This hybrid pipeline makes it possible to train localization capability at scale, even with limited annotated bounding boxes. Our results show that combining a small number of real annotations with pseudo-labels consistently improves IoU recall compared to real labels alone, highlighting the value of weak supervision.

Despite these promising results, there are clear limitations. First, our localization experiments were conducted on only 54 expert-annotated images focused on pneumonia and pneumothorax, which limits generalizability. Expanding the test set to include diverse pathologies, such as effusions, nodules, and fractures, will be essential for robust validation. Second, pseudo-labels generated by Grad-CAM are affected by its coarse resolution, which can lead to bounding boxes that partially miss or overshoot lesions. Incorporating improved methods like Grad-CAM++ or integrating anatomical priors (e.g., lung field masks) could yield more precise pseudo-labels. Third, our system outputs a single bounding box per image, yet many real-world cases have multiple concurrent findings. Extending the model architecture to handle multi-label outputs and overlapping lesions is an important next step. Another open challenge is the potential risk of false positives. A misleading bounding box, for instance, mistaking a normal apex for a pneumothorax, could misinform clinical decisions if the output is over-trusted. Introducing uncertainty quantification, ensemble predictions, or radiologist-in-the-loop validation could help mitigate these risks. Finally, while our work focuses on chest X-rays, the general framework could be adapted to other modalities like mammography or CT. However, extending current vision–language models to handle 3D volumetric data or sequential imaging remains technically challenging and warrants further research.

In future work, we plan to: (1) expand the pseudo-labeling pipeline to cover more conditions using multi-label classification for all CheXpert labels; (2) refine the pseudo-label quality with advanced saliency or segmentation-guided CAM methods; (3) validate the system in clinically realistic reader studies with radiologists; and (4) explore adapting the approach to other modalities and multi-slice imaging.

VI. Conclusion

In this work, we proposed a unified multimodal framework that jointly generates detailed radiology reports and localizes abnormalities in chest X-rays. By fine-tuning a large vision–language model (Qwen2.5-VL-7B) with limited expert annotations and Grad-CAM-based pseudo-labels, our approach demonstrates that weak supervision can effectively enhance lesion localization while preserving strong report generation quality. Experimental results on the MIMIC-CXR dataset show that combining real and pseudo labels significantly improves localization recall compared to using real labels alone. Meanwhile, our model's text generation remains competitive with state-of-the-art baselines while achieving superior inference efficiency. This study highlights the potential of large pretrained VLMs to deliver structured, interpretable outputs that align textual findings with visual evidence, a key requirement for trustworthy medical AI. Future work will expand pseudo-labeling to cover more pathologies, improve bounding box precision with advanced saliency methods, support multiple lesion outputs per image, and validate the system's clinical utility through real-world reader studies.

Acknowledgment

This work was partially supported by the Office of Research and Sponsored Programs and the Integrative Science and Technology Fellowship 2025, Kean University.

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